

Currents, Residues, and the Bochner–Martinelli Kernel

Andrea Taccani

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Abstract

These notes develop the theory of currents as an enlargement of differential forms. Currents contain locally integrable forms, geometric chains, singular kernels, and residue terms, while the current complexes compute the same de Rham and Dolbeault cohomology as smooth forms. The Bochner–Martinelli kernel is treated as the higher-dimensional Cauchy kernel: it is obtained from the real spherical kernel, its residue is computed, and its relation with the one-variable Cauchy kernel is explained on the blow-up of $\mathbb{C}^n$ at the origin. The local $\bar{\partial}$ -homotopy formula is formulated through the current identity on the diagonal, which fixes the signs without choosing ad hoc Koppelman components.

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1 Position in the Oberseminar

The Oberseminar moves through the chain

$$\begin{aligned}
&\text{Beilinson–Deligne cohomology} \rightsquigarrow \text{distributions} \rightsquigarrow \text{currents} \\
&\rightsquigarrow \text{tempered currents} \\
&\rightsquigarrow \text{regulators and Beilinson-type conjectures.}
\end{aligned}$$

The present report supplies the current-theoretic part. A distribution is a continuous functional on compactly supported test functions. A current is the corresponding notion with compactly supported test forms. This permits smooth forms, integration over chains, singular kernels, and residue terms to live in one differential complex.

The two cohomological statements proved below are

$$H_{\text{dR}}^*(M; \mathbb{C}) \simeq H^*(\mathcal{D}^*(M), d)$$

for a smooth real manifold M , and

$$H^q(X, \Omega_X^p) \simeq H^q(\mathcal{D}^{p,*}(X), \bar{\partial})$$

for a complex manifold X .

2 Overview

We first recall distributions and define currents with a cohomological degree convention. The current differential is the transpose of the differential on test forms, with the sign chosen so that a smooth form and its associated current have the same differential. We then isolate the residue mechanism in one complex variable and in the real spherical kernel. The $(n, n-1)$ component gives the Bochner–Martinelli kernel.

For the $\bar{\partial}$ -homotopy formula, the efficient object is not a list of separately signed Koppelman components. If

$$\delta : \mathbb{C}_z^n \times \mathbb{C}_w^n \longrightarrow \mathbb{C}^n, \quad \delta(z, w) = w - z,$$

then the pullback of the Bochner–Martinelli current satisfies

$$\bar{\partial} \delta^* T_\beta = [\Delta]$$

on $\mathbb{C}_z^n \times \mathbb{C}_w^n$, where Δ is the diagonal. Fiber integration against this current gives the homotopy operator. This formulation fixes all Koszul signs through the ordinary Leibniz rule and avoids the sign ambiguity that occurs if the mixed $d\bar{z}$ and $d\bar{w}$ factors are regrouped without an explicit convention.

3 From distributions to currents

3.1 Distributions

Let $U \subseteq \mathbb{R}^n$ be open. Write $C_c^\infty(U)$ for the space of compactly supported smooth functions with its standard test-function topology.

Definition 3.1. A distribution on U is a continuous linear functional

$$T : C_c^\infty(U) \longrightarrow \mathbb{C}.$$

If $\psi \in L_{\text{loc}}^1(U)$, then

$$T_\psi(\varphi) = \int_U \psi(x)\varphi(x) dx$$

defines a distribution. The delta distribution at $a \in U$ is $\delta_a(\varphi) = \varphi(a)$.

If $D_i = \partial/\partial x_i$, the distributional derivative is

$$(D_i T)(\varphi) = -T(D_i \varphi).$$

If ψ is smooth, integration by parts gives $D_i T_\psi = T_{D_i \psi}$.

Example 3.2 (Heaviside function). Let H be 0 on $(-\infty, 0)$ and 1 on $(0, \infty)$. Then

$$DT_H = \delta_0.$$

Indeed,

$$(DT_H)(\varphi) = - \int_0^\infty \varphi'(x) dx = \varphi(0).$$

The distributional derivative records the boundary term concentrated at the jump.

3.2 Definition of currents

Let M be an oriented smooth manifold of real dimension m , and let $\mathcal{A}_c^k(M)$ denote compactly supported smooth k -forms.

Definition 3.3 (Cohomological convention). A current of degree q is a continuous linear functional

$$T : \mathcal{A}_c^{m-q}(M) \longrightarrow \mathbb{C}.$$

The space of such currents is denoted $\mathcal{D}^q(M)$.

A smooth q -form ω defines

$$T_\omega(\eta) = \int_M \omega \wedge \eta, \quad \eta \in \mathcal{A}_c^{m-q}(M).$$

More generally, every locally integrable q -form defines a current by the same formula.

If Γ is an oriented piecewise smooth $(m - q)$ -chain, then

$$T_\Gamma(\eta) = \int_\Gamma \eta$$

defines a degree- q current. In complex geometry, an analytic subset of codimension p defines a current of type (p, p) .

3.3 Local description

On a coordinate open set $U \subseteq \mathbb{R}^m$, every degree- q current has a unique expression

$$T = \sum_{|I|=q} T_I dx_I$$

with scalar distribution coefficients T_I . This follows by testing against functions times the complementary coordinate forms. Conversely, such a finite sum defines a current. Thus currents are differential forms with distribution coefficients, although the intrinsic definition is the functional on compactly supported test forms.

4 The differential on currents

Let $T \in \mathcal{D}^q(M)$. Define $dT \in \mathcal{D}^{q+1}(M)$ by

$$(dT)(\eta) = (-1)^{q+1}T(d\eta), \quad \eta \in \mathcal{A}_c^{m-q-1}(M).$$

Proposition 4.1 (Compatibility with smooth forms). *For every smooth q -form ω ,*

$$dT_\omega = T_{d\omega}.$$

Proof. For $\eta \in \mathcal{A}_c^{m-q-1}(M)$, Stokes' theorem applied to the compactly supported form $\omega \wedge \eta$ gives

$$0 = \int_M d(\omega \wedge \eta) = \int_M d\omega \wedge \eta + (-1)^q \int_M \omega \wedge d\eta.$$

Hence

$$(dT_\omega)(\eta) = (-1)^{q+1} \int_M \omega \wedge d\eta = \int_M d\omega \wedge \eta.$$

□

Proposition 4.2 (Compatibility with boundaries). *If Γ is an oriented $(m-q)$ -chain, then*

$$dT_\Gamma = (-1)^{q+1}T_{\partial\Gamma}.$$

Proof. For $\eta \in \mathcal{A}_c^{m-q-1}(M)$,

$$(dT_\Gamma)(\eta) = (-1)^{q+1} \int_\Gamma d\eta = (-1)^{q+1} \int_{\partial\Gamma} \eta.$$

□

5 Residues and the Cauchy kernel

5.1 The residue pattern

Let ω be a locally integrable q -form, smooth on $M \setminus S$, where S is closed. Assume that the ordinary form $d\omega$ on $M \setminus S$ extends to a locally integrable $(q+1)$ -form on M . The residue current is

$$R(\omega) := dT_\omega - T_{d\omega}.$$

It is supported on S . Indeed, on any open set disjoint from S , Proposition 4.1 applies to the smooth form ω and the two terms agree. Thus the residue measures the contribution lost by differentiating only on the nonsingular locus.

5.2 Complex type convention

Let X be a complex manifold of complex dimension n . Write $\mathcal{A}^{p,q}(X)$ for smooth forms of type (p,q) and $\mathcal{A}_c^{p,q}(X)$ for compactly supported ones. A current of type (p,q) is a continuous functional

$$T : \mathcal{A}_c^{n-p,n-q}(X) \longrightarrow \mathbb{C}.$$

The space is denoted $\mathcal{D}^{p,q}(X)$.

For $T \in \mathcal{D}^{p,q}(X)$ define

$$(\bar{\partial}T)(\eta) = (-1)^{p+q+1}T(\bar{\partial}\eta).$$

With this convention, $\bar{\partial}T_\omega = T_{\bar{\partial}\omega}$ for every smooth (p,q) -form ω .

5.3 The Cauchy residue

On $\mathbb{C}$ set

$$\kappa = \frac{1}{2\pi i} \frac{dz}{z}.$$

It is locally integrable, smooth on $\mathbb{C}^*$, and $\bar{\partial}\kappa = 0$ on $\mathbb{C}^*$.

Proposition 5.1 (Cauchy residue). *As currents on $\mathbb{C}$,*

$$\bar{\partial}T_\kappa = \delta_0.$$

Proof. Let $f \in C_c^\infty(\mathbb{C})$ and choose R with $\text{supp}(f) \subset B_R(0)$. Since κ has type $(1, 0)$,

$$(\bar{\partial}T_\kappa)(f) = \lim_{\varepsilon \rightarrow 0} \int_{B_R \setminus B_\varepsilon} \kappa \wedge \bar{\partial}f.$$

On $\mathbb{C}^*$, $\bar{\partial}(f\kappa) = \bar{\partial}f \wedge \kappa$, and $\kappa \wedge \bar{\partial}f = -\bar{\partial}f \wedge \kappa$. Stokes' theorem on $B_R \setminus B_\varepsilon$ therefore yields

$$(\bar{\partial}T_\kappa)(f) = \lim_{\varepsilon \rightarrow 0} \int_{|z|=\varepsilon} f(z) \frac{1}{2\pi i} \frac{dz}{z}.$$

Writing $z = \varepsilon e^{i\theta}$ gives

$$\frac{1}{2\pi} \int_0^{2\pi} f(\varepsilon e^{i\theta}) d\theta \longrightarrow f(0).$$

□

6 The real spherical kernel

Let $x = (x_1, \dots, x_m)$ be the standard coordinates on $\mathbb{R}^m$, $m \geq 2$, let

$$r = \|x\|, \quad \Phi = dx_1 \wedge \dots \wedge dx_m,$$

and let

$$R = \sum_{j=1}^m x_j \frac{\partial}{\partial x_j}$$

be the radial vector field. Define

$$\alpha_m = c_m \frac{\iota_R \Phi}{r^m}, \quad c_m = \frac{1}{\text{vol}(S^{m-1})}.$$

Proposition 6.1 (Real spherical residue). *The form α_m is locally integrable on $\mathbb{R}^m$, smooth and closed on $\mathbb{R}^m \setminus \{0\}$, and*

$$dT_{\alpha_m} = \delta_0.$$

Proof. The coefficients have size $O(r^{1-m})$, which is locally integrable because the polar volume element is $r^{m-1} dr d\sigma$.

Let

$$\pi : \mathbb{R}^m \setminus \{0\} \longrightarrow S^{m-1}, \quad \pi(x) = x/\|x\|.$$

In polar coordinates,

$$\Phi = r^{m-1} dr \wedge \pi^* \omega_{S^{m-1}}, \quad R = r \frac{\partial}{\partial r}.$$

Hence

$$\iota_R \Phi = r^m \pi^* \omega_{S^{m-1}}, \quad \alpha_m = c_m \pi^* \omega_{S^{m-1}}.$$

Thus $d\alpha_m = 0$ away from 0 and $\int_{\partial B_\varepsilon} \alpha_m = 1$.

For $f \in C_c^\infty(\mathbb{R}^m)$, apply Stokes' theorem on $B_{R_0} \setminus B_\varepsilon$ with R_0 larger than the support of f . The outer boundary term vanishes, and the current sign gives

$$(dT_{\alpha_m})(f) = \lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon} f \alpha_m = f(0).$$

□

The form α_m is the normalized angular form around the origin. The Bochner–Martinelli kernel is its complex component of the bidegree appropriate for $\bar{\partial}$.

7 The Bochner–Martinelli kernel

7.1 Construction and residue

On $\mathbb{C}^n$ write $z_j = x_j + iy_j$ and

$$r^2 = \|z\|^2 = \sum_{j=1}^n |z_j|^2, \quad \Omega = dz_1 \wedge \cdots \wedge dz_n, \quad \bar{\Omega} = d\bar{z}_1 \wedge \cdots \wedge d\bar{z}_n.$$

Let

$$W = \sum_{j=1}^n \bar{z}_j \frac{\partial}{\partial \bar{z}_j}.$$

Define

$$\beta(z) = C_n \frac{\Omega \wedge \iota_W \bar{\Omega}}{\|z\|^{2n}}$$

with C_n chosen so that

$$\int_{\partial B_\varepsilon(0)} \beta = 1$$

for the complex boundary orientation. Equivalently,

$$\beta(z) = C_n \sum_{j=1}^n (-1)^{j-1} \frac{\bar{z}_j dz_1 \wedge \cdots \wedge dz_n \wedge d\bar{z}_1 \wedge \cdots \wedge \widehat{d\bar{z}_j} \wedge \cdots \wedge d\bar{z}_n}{\|z\|^{2n}}.$$

Proposition 7.1 (Bochner–Martinelli residue). *The form β is locally integrable, smooth on $\mathbb{C}^n \setminus \{0\}$, satisfies*

$$\bar{\partial}\beta = 0 \quad \text{on } \mathbb{C}^n \setminus \{0\},$$

and as a current

$$\bar{\partial}T_\beta = \delta_0.$$

Proof. The numerator is $O(r)$ and the denominator is r^{2n} , so the coefficients are $O(r^{1-2n})$ and therefore locally integrable in real dimension $2n$.

Set

$$A = \iota_W \bar{\Omega} = \sum_{j=1}^n (-1)^{j-1} \bar{z}_j d\bar{z}_1 \wedge \cdots \wedge \widehat{d\bar{z}_j} \wedge \cdots \wedge d\bar{z}_n.$$

Then $\bar{\partial}A = n\bar{\Omega}$ and

$$\bar{\partial}(r^{-2n}) = -nr^{-2n-2} \sum_{k=1}^n z_k d\bar{z}_k.$$

Moreover

$$\left(\sum_{k=1}^n z_k d\bar{z}_k \right) \wedge A = r^2 \bar{\Omega}.$$

Consequently $\bar{\partial}(r^{-2n}A) = 0$ on $\mathbb{C}^n \setminus \{0\}$, and hence $\bar{\partial}\beta = 0$ there.

The current residue is obtained exactly as in Proposition 5.1. For $f \in C_c^\infty(\mathbb{C}^n)$, Stokes' theorem on a punctured ball and the normalization of C_n give

$$(\bar{\partial}T_\beta)(f) = \lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon} f\beta = f(0).$$

□

7.2 Bochner–Martinelli formula for functions

Proposition 7.2. *Let $B_R = B_R(0) \subset \mathbb{C}^n$ and $f \in C^\infty(\overline{B_R})$. Then*

$$f(0) = \int_{\partial B_R} f\beta - \int_{B_R} \bar{\partial}f \wedge \beta.$$

If f is holomorphic, the second term vanishes.

Proof. On $B_R \setminus B_\varepsilon$ one has $\bar{\partial}\beta = 0$, and since $f\beta$ has type $(n, n-1)$,

$$d(f\beta) = \bar{\partial}(f\beta) = \bar{\partial}f \wedge \beta.$$

Thus

$$\int_{B_R \setminus B_\varepsilon} \bar{\partial}f \wedge \beta = \int_{\partial B_R} f\beta - \int_{\partial B_\varepsilon} f\beta.$$

The final integral tends to $f(0)$.

□

7.3 Blow-up interpretation

Let

$$\rho : \mathbb{C}^n \setminus \{0\} \rightarrow \mathbb{P}^{n-1}, \quad z \mapsto [z],$$

and let

$$\mu : \text{Bl}_0 \mathbb{C}^n \rightarrow \mathbb{C}^n$$

be the blow-up. Its exceptional divisor is $E \simeq \mathbb{P}^{n-1}$, and the blow-up is the total space of $\mathcal{O}_{\mathbb{P}^{n-1}}(-1)$. Let

$$p : \text{Bl}_0 \mathbb{C}^n \rightarrow \mathbb{P}^{n-1}$$

be the bundle projection.

Let ω_{FS} be normalized by

$$\rho^* \omega_{\text{FS}} = \frac{i}{2\pi} \partial \bar{\partial} \log \|z\|^2.$$

Up to the corresponding normalization constant,

$$\beta = C'_n \partial \log \|z\|^2 \wedge (\partial \bar{\partial} \log \|z\|^2)^{n-1}.$$

Proposition 7.3 (Cauchy factor on the blow-up). *On the affine blow-up chart*

$$z_1 = \lambda, \quad z_j = \lambda u_j \quad (j \geq 2),$$

one has on $\lambda \neq 0$

$$\mu^* \beta = C''_n \frac{d\lambda}{\lambda} \wedge p^*(\omega_{\text{FS}}^{n-1}).$$

Thus the Bochner–Martinelli kernel becomes the one-variable logarithmic kernel in the tautological line direction, multiplied by the Fubini–Study volume form in the projective directions.

Proof. Set $\theta = \partial \log \|z\|^2$. Both

$$\theta \wedge (\bar{\partial}\theta)^{n-1} \quad \text{and} \quad \frac{\Omega \wedge \iota_W \bar{\Omega}}{\|z\|^{2n}}$$

are $U(n)$ -invariant $(n, n-1)$ -forms with the same homogeneity. Evaluating at $(r, 0, \dots, 0)$ shows that they differ by a nonzero constant.

On the blow-up chart,

$$\|z\|^2 = |\lambda|^2(1 + |u_2|^2 + \dots + |u_n|^2).$$

Hence, away from E ,

$$\partial \log \|z\|^2 = \frac{d\lambda}{\lambda} + \partial \log(1 + |u|^2),$$

while

$$\partial \bar{\partial} \log \|z\|^2 = \partial \bar{\partial} \log(1 + |u|^2).$$

The second term in the first display wedges to zero with the $(n-1)$ st power of the Fubini–Study form because it would have holomorphic degree n in the $n-1$ projective variables. Only $d\lambda/\lambda$ remains. $\square$

8 The local $\bar{\partial}$ -homotopy formula

8.1 Regularity in degree zero

Theorem 8.1 (Holomorphic distributions). *Let $U \subseteq \mathbb{C}^n$ be open. If $T \in \mathcal{D}^{0,0}(U)$ and $\bar{\partial}T = 0$, then there is a unique $f \in \mathcal{O}(U)$ such that $T = T_f$.*

Proof. Locally identify T with a scalar distribution. The equation $\bar{\partial}T = 0$ means

$$\frac{\partial T}{\partial \bar{z}_j} = 0 \quad (j = 1, \dots, n)$$

in the distributional sense. Therefore

$$\Delta T = 4 \sum_{j=1}^n \frac{\partial^2 T}{\partial z_j \partial \bar{z}_j} = 0.$$

Weyl’s lemma implies that T is represented by a smooth harmonic function f . The distributional Cauchy–Riemann equations then become the ordinary equations $\partial f / \partial \bar{z}_j = 0$, so f is holomorphic. Uniqueness follows because a holomorphic function defining the zero distribution is identically zero. $\square$

8.2 The diagonal current and the Koppelman operator

Let

$$\delta : \mathbb{C}_z^n \times \mathbb{C}_w^n \rightarrow \mathbb{C}^n, \quad \delta(z, w) = w - z,$$

and let $B = \delta^* T_\beta$. Since δ is a submersion and $\delta^{-1}(0) = \Delta$, the complex orientation on the diagonal gives

$$\bar{\partial}B = [\Delta] \tag{8.1}$$

as currents on $\mathbb{C}_z^n \times \mathbb{C}_w^n$. This is the translated Bochner–Martinelli residue identity.

Let π_z and π_w denote the two projections. If $\varphi \in \mathcal{A}_c^{p,q}(\mathbb{C}^n)$ with $q > 0$, define

$$K\varphi := (\pi_z)_*(B \wedge \pi_w^* \varphi). \tag{8.2}$$

The product is locally integrable because B has a singularity of order $O(|w - z|^{1-2n})$, and φ is smooth. The pushforward is well-defined because φ has compact support in the w -variable. Bidegree shows that $K\varphi$ has type $(p, q - 1)$.

The advantage of (8.2) is that the mixed $d\bar{z}$ and $d\bar{w}$ terms are not reordered by hand. Fiber integration performs the required component extraction with one fixed orientation convention.

Theorem 8.2 (Bochner–Martinelli–Koppelman homotopy). *For every $\varphi \in \mathcal{A}_c^{p,q}(\mathbb{C}^n)$ with $q > 0$,*

$$\bar{\partial}K\varphi + K\bar{\partial}\varphi = \varphi. \quad (8.3)$$

Proof. The fiber of π_z has real dimension $2n$, hence fiber integration commutes with the exterior differential, and therefore with the $(0, 1)$ component $\bar{\partial}$, for forms with proper support over the base. Since B has total degree $2n - 1$, the Leibniz rule gives

$$\bar{\partial}(B \wedge \pi_w^* \varphi) = (\bar{\partial}B) \wedge \pi_w^* \varphi - B \wedge \pi_w^* (\bar{\partial}\varphi).$$

Using (8.1) and pushing forward,

$$\begin{aligned} \bar{\partial}K\varphi &= (\pi_z)_*([\Delta] \wedge \pi_w^* \varphi) - (\pi_z)_*(B \wedge \pi_w^* (\bar{\partial}\varphi)) \\ &= \varphi - K\bar{\partial}\varphi. \end{aligned}$$

The first term is φ because $\pi_z|_{\Delta} : \Delta \rightarrow \mathbb{C}^n$ is the identity and $\pi_w|_{\Delta} = \pi_z|_{\Delta}$. $\square$

Remark 8.3 (Relation with component formulas). Expanding $d\overline{(w - z)}_j = d\bar{w}_j - d\bar{z}_j$ decomposes B into the usual Koppelman components. If these components are regrouped as a tensor product of z -forms and w -forms, the relation away from the diagonal contains degree-dependent Koszul signs. Formula (8.3) is the same Koppelman formula, but (8.1)–(8.2) keep those signs inside the standard calculus of currents and fiber integration. Compare [6, Chapter I, §3, especially the Koppelman formula and the Dolbeault–Grothendieck lemma].

8.3 Extension to compactly supported currents

For compactly supported currents, the operator K is defined by distributional convolution with the Bochner–Martinelli current, equivalently by transposition from the kernel in (8.2). Convolution is defined because one factor has compact support. It agrees with (8.2) on smooth compactly supported forms.

Proposition 8.4 (Current homotopy formula). *If $T \in \mathcal{D}^{p,q}(\mathbb{C}^n)$ has compact support and $q > 0$, then*

$$\bar{\partial}KT + K\bar{\partial}T = T \quad (8.4)$$

in $\mathcal{D}^{p,q}(\mathbb{C}^n)$.

Proof. Choose a standard smoothing kernel ρ_ε and let $T_\varepsilon = T * \rho_\varepsilon$. Then T_ε is smooth, its support stays in a fixed compact set for small ε , and

$$T_\varepsilon \rightarrow T, \quad \bar{\partial}T_\varepsilon = (\bar{\partial}T) * \rho_\varepsilon \rightarrow \bar{\partial}T$$

in the topology of currents. The kernel operator K is continuous on compactly supported distributions, since it is convolution with a fixed distribution. Applying (8.3) to T_ε and passing to the limit gives (8.4). This is the standard extension of the Koppelman formula to currents; see [6, Chapter I, §3]. $\square$

8.4 Local exactness for currents

Corollary 8.5 (Local $\bar{\partial}$ -Poincare lemma for currents). *Let $U \subseteq \mathbb{C}^n$ be open, $q > 0$, and $T \in \mathcal{D}^{p,q}(U)$ with $\bar{\partial}T = 0$. For every $a \in U$ there are an open neighborhood V of a and $S \in \mathcal{D}^{p,q-1}(V)$ such that*

$$T|_V = \bar{\partial}S.$$

Proof. Choose relatively compact balls

$$a \in V \Subset W \Subset U$$

and a cutoff $\chi \in C_c^\infty(W)$ equal to 1 on a neighborhood of $\bar{V}$. Extend χT by zero to $\mathbb{C}^n$. Since $\bar{\partial}T = 0$,

$$\bar{\partial}(\chi T) = (\bar{\partial}\chi) \wedge T.$$

The global homotopy formula (8.4) gives

$$\chi T = \bar{\partial}K(\chi T) + K((\bar{\partial}\chi) \wedge T). \quad (8.5)$$

Set

$$E := K((\bar{\partial}\chi) \wedge T).$$

The support of $(\bar{\partial}\chi) \wedge T$ is disjoint from V . For $z \in V$, the kernel is therefore smooth in a neighborhood of that support; pairing it with the fixed current produces a smooth (p, q) -form E on V . Restricting (8.5) to V gives

$$T = \bar{\partial}K(\chi T) + E.$$

Applying $\bar{\partial}$ shows $\bar{\partial}E = 0$. By the ordinary smooth $\bar{\partial}$ -Poincare lemma, after shrinking V if necessary there is a smooth $(p, q-1)$ -form R on V with $E = \bar{\partial}R$. Thus

$$T = \bar{\partial}(K(\chi T) + R)$$

on V . □

9 Cohomology with currents

9.1 Dolbeault current sheaves

Let X be a complex manifold of dimension n . For $U \subseteq X$ open, set

$$\mathcal{D}_X^{p,q}(U) := (\mathcal{A}_c^{n-p, n-q}(U))'.$$

If $V \subseteq U$ and $T \in \mathcal{D}_X^{p,q}(U)$, define $T|_V$ by testing T against the extension by zero to U of a compactly supported test form on V . This extension is smooth because its support is compactly contained in V .

Proposition 9.1 (Dolbeault current resolution). *For each (p, q) , $U \mapsto \mathcal{D}_X^{p,q}(U)$ is a sheaf. Moreover*

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{D}_X^{p,0} \xrightarrow{\bar{\partial}} \mathcal{D}_X^{p,1} \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \mathcal{D}_X^{p,n} \longrightarrow 0 \quad (9.1)$$

is a fine resolution of Ω_X^p .

Proof. For the sheaf property, let $U = \bigcup_i U_i$. Uniqueness follows by taking a finite smooth partition of unity near the compact support of a test form and reducing its evaluation to the restrictions on the U_i . Compatible local currents glue by the same formula

$$T(\eta) = \sum_i T_i(\rho_i \eta),$$

and independence of the partition follows after passing to a common refinement.

A holomorphic p -form α defines a $(p, 0)$ -current by integration. This map is injective: locally write $\alpha = \sum_I f_I dz_I$ and test against complementary forms times arbitrary compactly supported scalar functions.

If $T \in \mathcal{D}_X^{p,0}$ and $\bar{\partial}T = 0$, then locally

$$T = \sum_{|I|=p} T_I dz_I$$

and every scalar coefficient satisfies the distributional Cauchy–Riemann equations. Theorem 8.1 implies that every T_I is holomorphic. Thus the kernel at degree 0 is Ω_X^p .

Exactness in degree $q > 0$ is Corollary 8.5. Finally, $\mathcal{D}_X^{p,q}$ is a module over the sheaf of smooth functions. If $\{\rho_i\}$ is a smooth partition of unity subordinate to a locally finite cover, the endomorphisms $T \mapsto \rho_i T$ have the required supports and sum locally finitely to the identity. Hence each $\mathcal{D}_X^{p,q}$ is fine. $\square$

9.2 Dolbeault cohomology computed by currents

Theorem 9.2. *For every complex manifold X and every p, q ,*

$$H^q(X, \Omega_X^p) \simeq H^q(\Gamma(X, \mathcal{D}_X^{p,*}), \bar{\partial}). \quad (9.2)$$

Equivalently,

$$H^q(X, \Omega_X^p) \simeq \frac{\ker(\bar{\partial} : \mathcal{D}^{p,q}(X) \rightarrow \mathcal{D}^{p,q+1}(X))}{\operatorname{im}(\bar{\partial} : \mathcal{D}^{p,q-1}(X) \rightarrow \mathcal{D}^{p,q}(X))}.$$

Proof. Proposition 9.1 gives a fine, hence acyclic, resolution of Ω_X^p . The standard theorem on acyclic resolutions computes the sheaf cohomology of Ω_X^p by the cohomology of its complex of global sections. $\square$

Remark 9.3 (Comparison with smooth Dolbeault forms). The ordinary Dolbeault complex

$$0 \rightarrow \Omega_X^p \rightarrow \mathcal{A}_X^{p,0} \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \mathcal{A}_X^{p,n} \rightarrow 0$$

is also a fine resolution. The inclusion of smooth forms into currents is a morphism between these two resolutions inducing the identity on Ω_X^p . Therefore

$$\mathcal{A}_X^{p,*} \hookrightarrow \mathcal{D}_X^{p,*}$$

induces an isomorphism on global Dolbeault cohomology. Currents enlarge the class of representatives without changing the cohomology.

9.3 The de Rham current complex

Let M be a smooth real manifold of dimension m , and denote by $\mathcal{D}_M^q$ the sheaf of degree- q currents. The exterior derivative gives

$$\mathcal{D}_M^0 \xrightarrow{d} \mathcal{D}_M^1 \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{D}_M^m.$$

Proposition 9.4 (de Rham current resolution). *The sequence*

$$0 \longrightarrow \underline{\mathbb{C}}_M \longrightarrow \mathcal{D}_M^0 \xrightarrow{d} \mathcal{D}_M^1 \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{D}_M^m \longrightarrow 0 \quad (9.3)$$

is a fine resolution of the constant sheaf $\underline{\mathbb{C}}_M$.

Proof. The sheaf and fineness arguments are the real analogues of Proposition 9.1. It remains to prove local exactness.

In degree 0, a current is a scalar distribution. If $dT = 0$, every first distributional derivative vanishes. Convolving locally with a standard mollifier gives smooth functions with zero differential, hence locally constant functions. Passing to the limit shows that T is a locally constant distribution.

In positive degree, use the local regularization theorem for currents. On a coordinate ball B and a smaller ball $B' \Subset B$, de Rham regularization gives continuous operators

$$R_\varepsilon : \mathcal{D}^q(B) \longrightarrow \mathcal{A}^q(B'), \quad A_\varepsilon : \mathcal{D}^q(B) \longrightarrow \mathcal{D}^{q-1}(B')$$

such that $R_\varepsilon T$ is smooth and

$$R_\varepsilon T - T = dA_\varepsilon T + A_\varepsilon dT \quad (9.4)$$

on B' . Locally these operators are constructed from convolution with a mollifier and the homotopy between translation and the identity; see [6, Chapter I]. If $dT = 0$ and $q > 0$, then $R_\varepsilon T$ is a smooth closed q -form. The ordinary Poincaré lemma on a smaller ball gives a smooth $(q-1)$ -form η with $R_\varepsilon T = d\eta$. Equation (9.4) then gives

$$T = d(\eta - A_\varepsilon T).$$

Thus (9.3) is locally exact in positive degree. Since the current sheaves are fine, it is a fine resolution. $\square$

Corollary 9.5 (Cohomology comparison). *The inclusions of smooth forms into currents induce canonical isomorphisms*

$$H_{\text{dR}}^*(M; \mathbb{C}) \simeq H^*(\mathcal{D}^*(M), d)$$

and, for a complex manifold X ,

$$H^q(\mathcal{A}^{p,*}(X), \bar{\partial}) \simeq H^q(\mathcal{D}^{p,*}(X), \bar{\partial}) \simeq H^q(X, \Omega_X^p).$$

Proof. Proposition 9.4 identifies the current de Rham complex as a fine resolution of $\underline{\mathbb{C}}_M$, while the usual de Rham complex is another fine resolution of the same sheaf. The de Rham theorem identifies their common cohomology with $H_{\text{dR}}^*(M; \mathbb{C})$. The Dolbeault statement follows from Theorem 9.2 and the ordinary Dolbeault theorem. $\square$

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